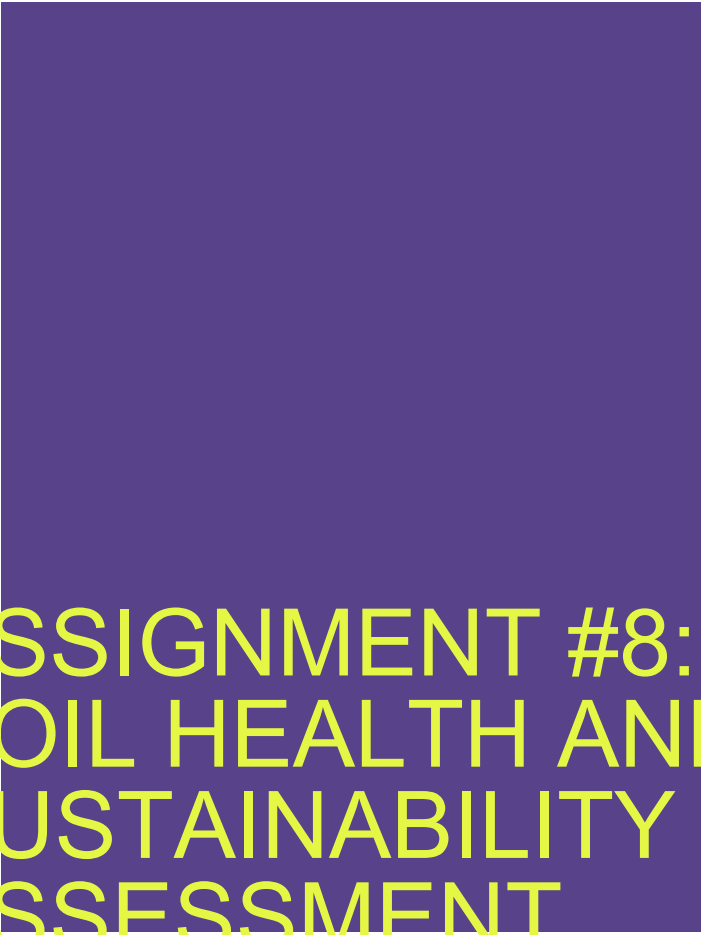




ASSIGNMENT #8: SOIL HEALTH AND SUSTAINABILITY METRICS ASSESSMENT

ASSIGNMENT OVERVIEW

FIELD	VALUE
ASSIGNMENT NUMBER	8
TITLE	Soil Health and Sustainability Metrics Assessment
GRADE VALUE	12 points
DUE DATE	March 30, 2026, 11:59 PM PT

OBJECTIVES

After completing this assignment, you will be able to:

- Interpret NRCS SSURGO soil data to assess the physical and chemical health of your fields.
- Visualize soil variability (Organic Matter, pH, CEC) across a large field package.
- Calculate at least two sustainability metrics (e.g., Carbon Sequestration Potential or Erosion Risk).
- Communicate soil-based management restrictions to stakeholders using professional visualizations.

INSTRUCTIONS

Step 1: Create Your Feature Branch

- Open your terminal in VS Code and switch to a new branch:

```
git checkout -b feature/assignment-08-soil-health
```

Step 2: Extract Soil Health Indicators

Your data package contains detailed NRCS soil attributes.

1. **Create a new notebook:** `notebooks/08_soil_sustainability.ipynb`.
2. **Task:** Filter your soil dataset for key indicators: Organic Matter (OM), pH, and Cation Exchange Capacity (CEC).
3. **Task:** Group these by Field ID to see which fields have the highest "Soil Health Score" based on these raw values.

Mark complete:

Step 3: Analyze Sustainability Metrics

Sustainability in row crops often revolves around soil stability and carbon.

1. **Task:** Identify fields with high Erosion Risk based on the "K-Factor" (soil erodibility) or slope gradients provided in the NRCS data.
2. **Task:** Calculate a simple Carbon Storage Potential metric based on the depth of the topsoil and organic matter percentage.

Mark complete:

Step 4: Visualize Soil Variability

1. **Task:** Create a bar chart or box plot comparing Organic Matter % across your top 10 most productive fields vs. your bottom 10.
2. **Task:** Create a spatial plot showing the distribution of Soil Drainage Classes (e.g., Well Drained vs. Poorly Drained) across your field cluster.

Mark complete:

Step 5: Design Your Dashboard Soil Section

1. **Task:** Design 1-2 visualizations focused specifically on Soil Health and Sustainability for

your final dashboard.

2. **Suggestion:** A "Sustainability Scorecard" or a "Soil Nutrients Radar Chart" works well here.

3. **Task:** Save the final plots in
`output/dashboard_assets/soil_health_metrics.png`.

Mark complete:

Step 6:Update Your Project Tracker

Open `docs/project/assignment-01-tracker.md`.

- Document the most common soil limitation found in your 200 fields (e.g., "70% of fields are limited by high clay content and poor drainage").
- Log the sustainability metrics you chose to track.

Mark complete:

GRADING CRITERIA (12POINTS TOTAL)

Criterion	Points	Description
Soil Data Analysis	4	Successfully extracted and

		cleaned key soil health indicators (OM, pH, etc.).
Sustainability Metrics	3	Correctly calculated and interpreted at least 2 sustainability or risk metrics.
Visual Analysis	3	Created high-quality charts showing variability between field performance and soil health.
Dashboard Design	2	Designed a dedicated soil health section for the final dashboard layout.